

ALEXANDER TIAN

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EDUCATION

University of California, Berkeley	May 2028
B.S. Industrial Engineering & Operations Research	GPA: 4.0
Organizations: Machine Learning at Berkeley	
Honors & Awards: USACO Platinum Division (Dec. 2024)	

EXPERIENCES

Amazon Annapurna Labs , Machine Learning Engineer (Contract)	Feb. 2026 - Present
- Optimized video diffusion model inference for AWS Trainium 2 chips - Developed fused NKI kernels for attention and 3D-RoPE modules, performing roofline analyses and MFU enhancements	
Westchester Technology Group , Information Technology Intern	Apr. 2025 - June 2025
- Assisted with diagnosing and resolving software/hardware issues - Performed system administration with Microsoft Azure, and used NinjaOne for device management and IT automation	
Chess.com , Software Engineering Intern	July 2024 - Aug. 2025
- Contributed to the backend explanation software with analysis features and further integration with chess engines - Laid the groundwork for human-like, adaptable bots in Chess.com's proprietary engine, Torch - Collaborated with engineers to identify issues, enhance code stability, and produce scalable software - Gained hands-on experience with backend development in C++ and CI/CD workflows	

PROJECTS

Algorithmic Chess Engine	Aug. 2022 - Dec. 2025
- Developed a formerly top-30, 3600+ rated, superhuman chess engine of 1000+ engines worldwide in C++ - Competed in the Chess.com Computer Chess Championship (CCC) and Top Chess Engine Championship (TCEC) - Crafted an exhaustive search algorithm in C++ bolstered by neural networks trained on 5 billion generated data points - Optimized performance through statistical tests, profiling, SIMD, and Simultaneous Multi-Processing techniques - Downloaded over 3000 times on GitHub	
YOLO-v11 Mechanistic Interpretability	Nov. 2025 - Jan. 2026
- Discovered circuits in YOLO-v11 responsible for object detection through pixel-based optimization of activations - Developed a novel importance-weighting algorithm for automatically identifying kernels relevant to object classes - Experimented with receptive fields and feature sensitivity in various layers of YOLO-v11	
Neural Network Framework	Mar. 2025 - Jun. 2025
- A modular, sequential, CPU-based neural network framework written from scratch, utilizing only a linear algebra library - Supports components from convolutional to masked multi-headed self-attention layers - Implemented a GPT-2 styled decoder-only LLM, matching loss of PyTorch-based methods on the tiny-shakespeare dataset	
N-Body Simulator	Oct. 2024 - Jan. 2025
- Created a simulator of orbital mechanics and the N-body problem capable of simulating 100,000 objects - Utilized numerical integration, asymptotic algorithms, and parallel processing with Numba - Researched error reduction techniques for approximations and optimized gravitational force calculations	

RESEARCH

Berkeley Artificial Intelligence Research - Pallas Group , Undergraduate Researcher	Feb. 2026 - Present
- Optimizing Variational AutoEncoders for latent video diffusion models - Experimenting with sparse methods to improve diffusion language model performance	

SKILLS

Spoken Languages: English (Native), Mandarin (Fluent), Spanish (Proficient)	
Technical Languages: C, C++, Python, Java, Triton, MATLAB, RISC-V	
Tools: PyTorch, NumPy, Git, Linux	